AI5707 Lab 2 -- Process Reference (EVD / SVD Image Reconstruction)

Personal working reference only -- not a submission document. Report is written independently.

Notation used throughout

Symbol	Meaning
A	grayscale image, as an n×n (or m×n) matrix of pixel intensities, values in [0, 255]
Q, Λ	EVD of A: $A = Q \Lambda Q^{-1}$. Q's columns are eigenvectors, Λ is diagonal with the eigenvalues
U, Σ, V^t	SVD of A: $A = U \Sigma V^t$. U, V are orthogonal; Σ is diagonal with the (non-negative, real) singular values
k	number of components kept; all others are zeroed out before reconstructing
A_k	rank-k reconstruction using only the top-k eigenvalues/singular values
$E(k)$	Frobenius-norm reconstruction error: $E(k) = \|A - A_k\|_F$ = sqrt(sum of squared pixel differences)

Pipeline (both images)

> load PNG (RGB) → resize color image → convert to grayscale matrix A → decompose (EVD and/or SVD) → zero out all but the top k eigen/singular values → reconstruct A_k → compare A vs A_k

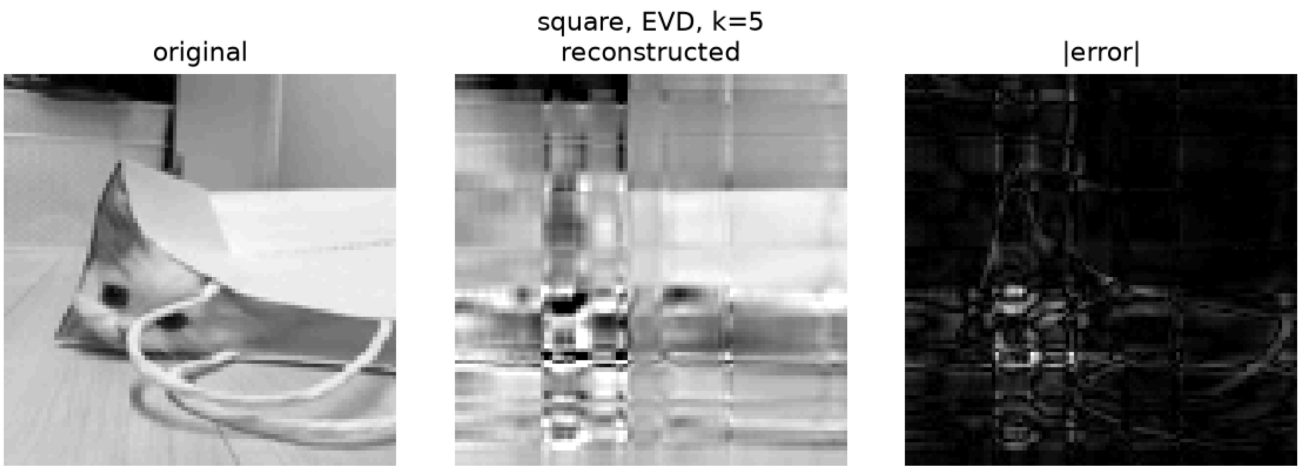
> Square image resized to exactly 100×100. Rectangular image resized so its longer side becomes 100, aspect ratio preserved (actual result: 520×600 → 87×100). Resize happens on the color image, before grayscale conversion.

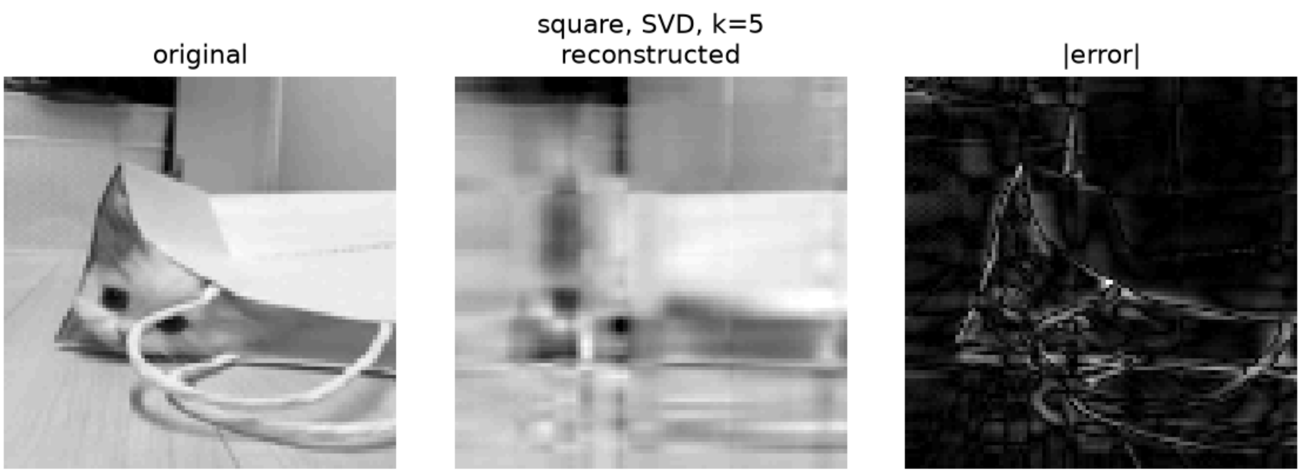
Task 1 -- Square image (EVD + SVD)

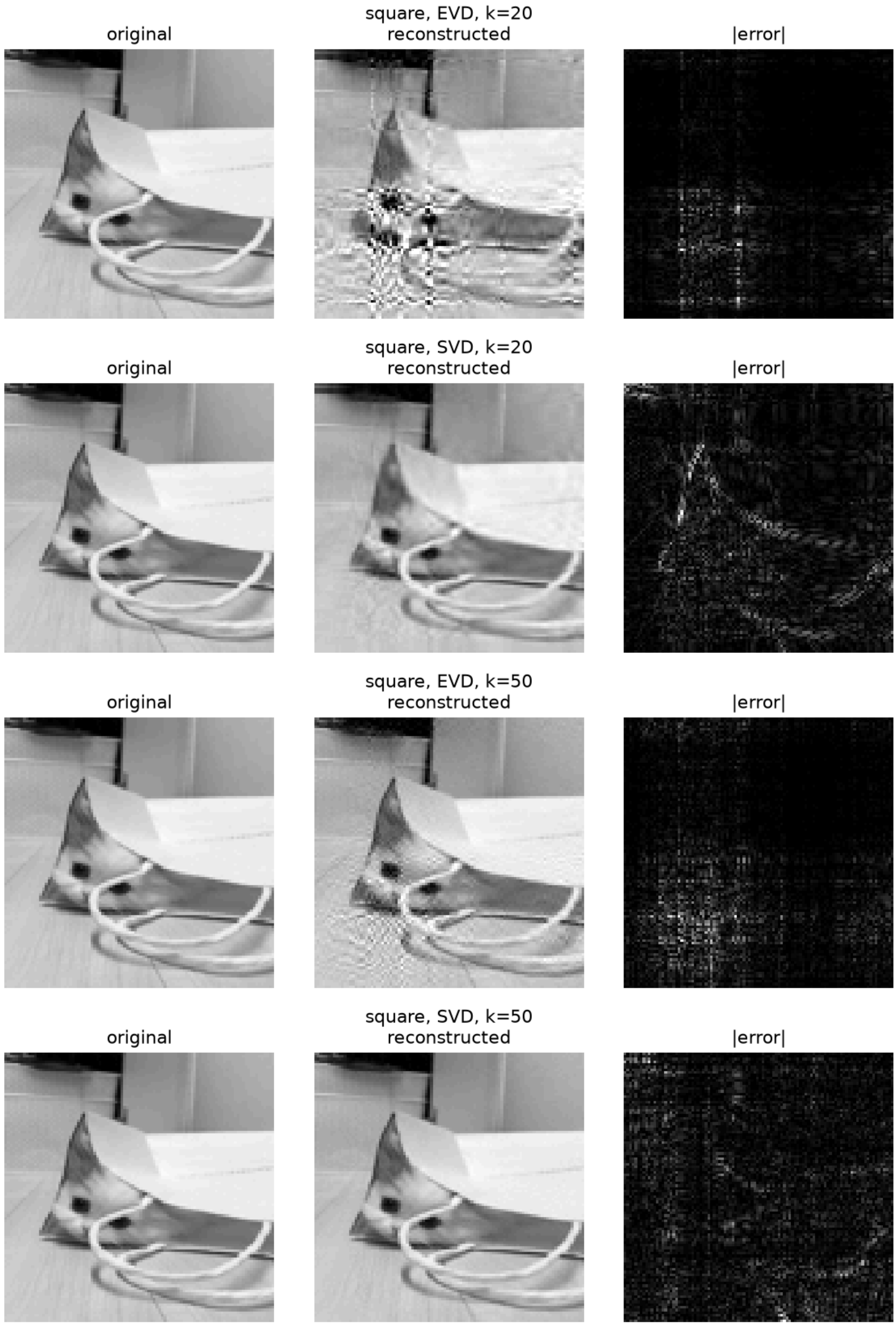
Load → resize → grayscale

A is 100×100, float values in [0, 255]

Reconstructions at k = 5, 20, 50

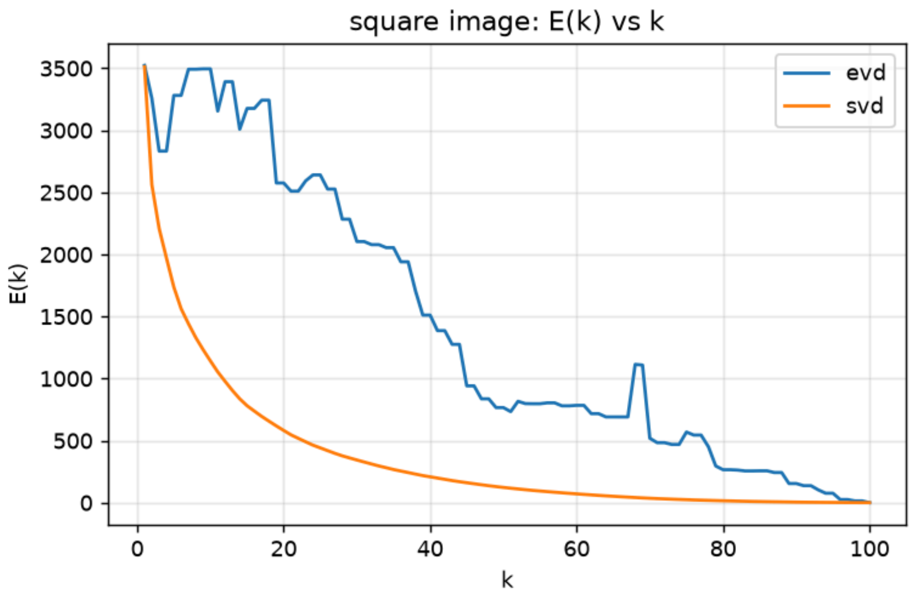





original
square, EVD, k=20
reconstructed
|error|
original
square, SVD, k=20
reconstructed
|error|
original
square, EVD, k=50
reconstructed
|error|
original
square, SVD, k=50
reconstructed
|error|

Reconstruction error

k	E(k) -- EVD	E(k) -- SVD
5	3281.02	1736.93
20	2575.44	581.84
50	765.98	122.67



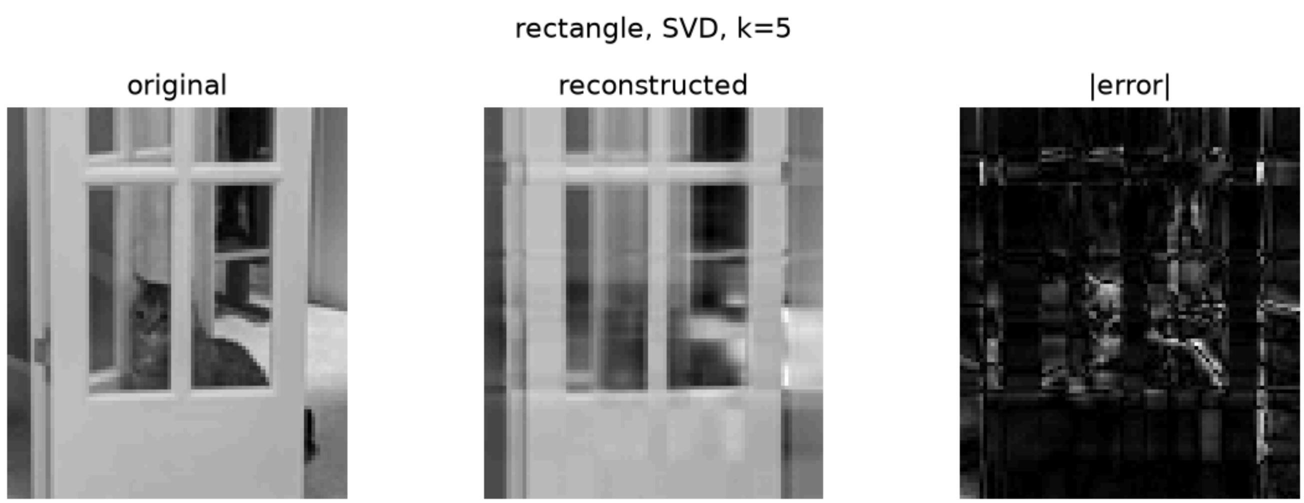
Task 2 -- Rectangular image (SVD only)

> No EVD here -- eigendecomposition is only defined for square matrices.

Load → resize (aspect-preserving) → grayscale

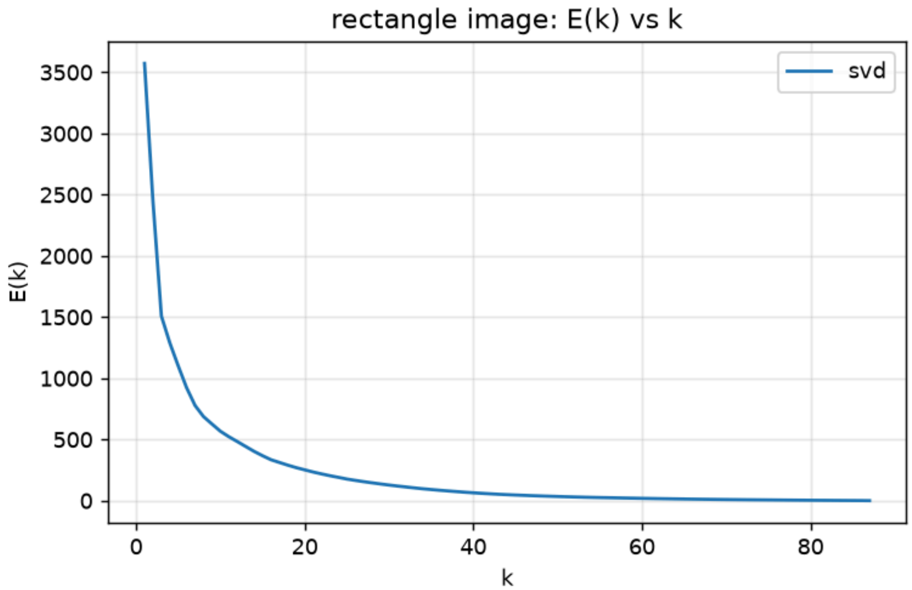
A is 87×100, float values in [0, 255]

Reconstructions at k = 5, 20, 50



Reconstruction error

k	E(k) -- SVD
5	1100.14
20	251.75
50	34.15


rectangle image: E(k) vs k
svd
E(k)
k
3500
3000
2500
2000
1500
1000
500
0
0
20
40
60
80